ASCIA OS

Next Generation Home Automation System

Product Specifications & Vision Document

Version 1.0 — Confidential — ASCIA Inc.

Introduction & Vision

ASCIA is a company specializing in advanced home automation, offering smart solutions for homes, warehouses, hotels and commercial spaces. Based on Home Assistant as its engine, ASCIA goes far beyond by offering its own home automation operating system — ASCIA OS — installed on a dedicated server, along with its own hardware equipment (PoE switches, door and window sensors, mmwave presence sensors, and other sensors such as vibration, temperature, air quality, humidity, vibration, water leak, thermal, etc., actuators, relay modules for LED lights, curtains, etc., switches with screen, scene switches, dimmer and normal switches, bulbs, spotlights, downlights and decorative lights, thermostats, tablet dashboards, cameras).

The intended customer experience is 100% Plug & Play: the customer plugs it in, turns it on, and the system does the rest.

MODULE 1 — System Architecture & Infrastructure

1.1 ASCIA OS — Home automation operating system

ASCIA OS is a fork of Home Assistant OS (HAOS) distributed as a disk image that can be installed on a mini-PC or dedicated server provided by ASCIA. It inherits all the capabilities of Home Assistant while adding a complete proprietary layer: interface, automations, network security, local AI, and hardware management.

- Based on Home Assistant Core + Supervisor + OS
- The frontend interface has been completely replaced by the ASCIA interface.
- Automatic startup of all services at boot

- Updates are managed via the ASCIA panel (manual, weekly, monthly or annual depending on customer choice)
- Compatible with the entire existing Home Assistant ecosystem (integrations, HACS, add-ons)

1.2 Plug & Play Deployment Model

The main objective is for a non-technical customer to be able to deploy a complete home automation system without the intervention of a network or IT professional.

Ideal installation process

- The customer receives: 1 pre-configured ASCIA server + N PoE switches + their ASCIA devices
- Step 1: Connect the modem → main switch → ASCIA server
- Step 2: Connect all devices to the switches
- Step 3: Access the ASCIA interface from your phone or browser
- Step 4: Initial guided setup (name, language, geographic area, creation of the 3D plan)
- Step 5: The devices appear automatically, ready to be configured.

At no point does the customer need to configure their modem, router, VLANs, or anything else at the network level.

1.3 Automatic isolated and secure network

From the first boot, ASCIA OS creates a secure and isolated network ecosystem without any external intervention.

Automatic network features

- Automatic creation of a dedicated VLAN for home automation devices
- Creation of an isolated WLAN (separate Wi-Fi network for IoT devices)
- Implementation of adaptive firewall rules protecting each device
- Automatic detection of new devices added to or removed from the network
- Automatic application of security rules on each new device detected
- Complete isolation of the main home network from IoT devices
- Continuous network monitoring with alerts in case of anomalies or intrusions
- Compatible with Omada (TP-Link) and UniFi (Ubiquiti) for advanced management

The customer does not need to make any changes to their existing modem or router.

MODULE 2 — Dynamic 3D/2D Interface

2.1 Real-time interactive 3D plan

At the heart of the ASCIA experience is a 3D/2D home dashboard where all devices, sensors, and controls are positioned in their actual locations. The visual rendering aims for the quality of SweetHome 3D with significantly improved fluidity and interactivity.

Plan creation

- Create directly within the ASCIA interface using the integrated editor and drag & drop.
- Importing a SweetHome 3D (.sh3d) or SketchUp (.skp) file — automatic conversion
- Scan the LiDAR data from the ASCIA mobile app (compatible with iPhone Pro and Android LiDAR) to automatically generate a 3D floor plan of the house.
- Switchable between 3D view and 2D view (floor plan)
- Option to lock or unlock the 3D model (lock/unlock navigation)

Real-time dynamism

- Open door → the door visually opens in 3D
- Blind lowered → the blind descends in 3D
- Light on → the light turns on in 3D with the actual intensity
- Red RGB light → the light renders the color red in 3D with colored light projection
- FP2 Presence Sensor (Aqara) → silhouettes appear in rooms and move in real time
- Water leak detected → affected area highlighted in red in the 3D view
- Vibration, pressure, robot vacuum cleaners, coffee machines → all dynamically represented
- Activateable heat map: color gradient based on temperature sensors in each zone

Interaction and configuration

- Click on a device in the 3D view → direct control (turn on, turn off, adjust)
- Drag and drop the devices onto the map to reposition them.
- Zone system: define a zone in 3D → the system automatically assigns it
- Automatically adds the device icon to its location on the map
- Automatic zoom on the affected area during an incident (e.g., water leak → zoom on valve)
- Configure presence sensors and other devices directly in 3D

2.2 Dynamic Emergency Dashboard

In the event of a critical incident (fire, water leak, intrusion), all dashboards automatically transform to display only the information and controls related to the incident.

- Water leak: zoom on the main valve in 3D + close-up of all water sensors + emergency number
- Fire: 3D plan with affected area in red + smoke/heat sensors + access points and exits

- Intrusion: full-screen view of cameras in the affected area + locks + history of recent movements

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MODULE 3 — Device Discovery & Integration

3.1 Universal Self-Discovery

As soon as a device is connected to the ASCIA network (cable or wireless), it is automatically detected, identified and integrated into the system without any code entry or complex manual configuration.

Supported protocols

- Zigbee (via Zigbee2MQTT) — complete visual interface
- Z-Wave
- Wi-Fi (MQTT, mDNS, SSDP, UPnP)
- PoE — automatic detection on the switch
- KNX — dedicated visual interface
- Bluetooth / BLE
- ONVIF / RTSP (IP cameras)
- Matter / Thread
- ASCIA Proprietary Products

Automated onboarding process

- Device detected → the system identifies its type (camera, sensor, light, relay, etc.)
- The system automatically assigns it a smart name based on its nature, the client's submission, and its area.
- The system requests: confirmation of name + assignment zone
- The device instantly appears in 3D in its assigned area with the correct icon, as well as in dedicated locations and automated settings.
- All associated entities are automatically created in the dashboard
- All groups are automatically created and modified according to the nature and zone of each device, for example (all bedroom lights, living room curtains, etc.).
- ASCIA PoE/Zigbee/Wi-Fi products → full profile automatically applied (camera mode, sensor mode, etc.)

3.2 Simplified Pairing

- 'Enable Zigbee pairing' button → all Zigbee devices within range appear
- The system asks for the type of device if it cannot be detected automatically.
- The system requests the zone of affiliation.

- The device is automatically placed in the 3D view and dashboard.
- The same applies to Wi-Fi, PoE, KNX, and Z-Wave.

3.3 Visual Interfaces by Protocol

Each protocol has its own simplified visual management panel, accessible without technical knowledge.

- KNX: visual mapping of group addresses, command testing, zone configuration
- Zigbee2MQTT: Network mesh map, device status, firmware updates, device configuration
- PoE/Wi-Fi: list of devices per switch/port, connection status, ping, signal quality
- Omada / Unifi: complete management of the home automation network (VLANs, APs, switches) from the ASCIA interface
- Global mapping interface: all devices, protocols, entities and their network state in a single view with visual mapping and diagrams on each link.

MODULE 4 — Security, Access & Confidentiality

4.1 Remote Access

- Cloud option: simplified access via ASCIA Cloud and multi-server management as well as support and remote diagnostics.
- VPN option: automatic configuration of a VPN (WireGuard, OpenVPN or tailscale) for 100% local and private access
- The customer activates the VPN option from the interface or mobile app — no network or phone configuration required
- VPN user management: creation, revocation, access duration
- ASCIA Support: Temporary access (15 min / 30 min / 1 hour) granted by the client for remote intervention

4.2 User & Permission Management

- Multi-user accounts: admin, family member, child, guest, service provider, support
- Child mode: no access to cameras, locks, or sensitive areas
- Guest mode: limited access to authorized areas, configurable duration
- Permissions by zone, by device, and by feature
- History of all connections: time, location, device used
- Recording of each connection attempt (successful or failed) with IP address and geolocation
- Two-factor authentication (2FA) available

4.3 Data Encryption & Security

- End-to-end encryption for remote communications
- Professional and automated management of encryption keys
- Self-renewing SSL/TLS certificates
- Weekly automated security audit with report

4.4 Privacy Mode

- Each zone, room or device can activate a 'Privacy' mode
- Private mode area → blurred area in 3D
- Global privacy mode → disables all cameras and microphones
- Fine-tuning: choose which devices are disabled in private mode
- History of privacy mode activations

MODULE 5 — Automation & Intelligence

5.1 Enhanced Traditional Automation

ASCIA OS retains and improves the Home Assistant automation engine while adding layers of intelligence and simplification.

- Visual automation editor (drag & drop of triggers, conditions, actions)
- Intelligent automation system based on the nature of the devices that self-adapts and facilitates the creation of automations.
- Library of pre-designed ASCIA blueprints for the most common use cases
- Automation timeline: see chronologically what was triggered and why
- Full log: each trigger recorded with the source device, the time, and the action performed

5.2 Automatic creation of automations and groups

The system automatically generates basic groups and automations as soon as devices are added, drastically reducing setup time.

- Add lights to 'Bedroom 1' → 'Bedroom 1 Lights' group created automatically
- Add a presence sensor in 'Living Room' → blueprint 'Living Room Presence → Living Room Lights' created and ready to activate
- Add blinds to an area → blind group created + blueprints for opening hours and weather suggested
- Automatically created automations are inactive by default — the client activates and customizes them.

- Scenes are created automatically according to pre-selected patterns (morning, evening, night, cinema, etc.).

5.3 AI for creating automations using natural language

The client can create complex automations simply by writing or speaking in natural language.

- Example: 'I want all the living room lights to turn on when I get home' → automation created immediately
- For complex automations, AI asks up to 10 clarifying questions for precise configuration.
- AI offers optimizations and variations of automation
- Available in text chat and voice commands
- AI learns from customer habits to anticipate their needs

5.4 Autopilot Mode

The system can be configured to operate completely autonomously, making decisions on behalf of the client according to predefined rules.

- The client (or ASCIA) configures the basic rules of the autopilot mode.
- The system decides which lights to turn on/off, when to arm the alarm, etc.
- This mode can always be modified and deactivated at any time.
- Machine learning based on history to refine decisions

5.5 Advanced Presence Simulation

- Replicating customer habits based on history (local machine learning)
- The lights, blinds, and appliances replicate typical usage patterns.
- More credible than classic random simulations
- Configurable: choose which devices participate in the simulation

5.6 Modes & Scenes

- Simplified mode creation: Morning, Evening, Cinema, Dinner, Away, Vacation, Night...
- The system suggests styles to the customer based on the season, time of day, and history.
- After learning, the system automatically activates the usual Tuesday evening or Sunday morning mode.
- Creating a mode = selecting devices + their desired state, without having to create automations manually

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MODULE 6 — Local Artificial Intelligence

6.1 Conversational AI Assistant

A local AI chatbot, trained on the home's data, answers questions and executes commands in natural language. Everything is processed locally — no data leaves the home.

- 'Is the oven on?' → 'Yes, the oven has been on for 23 minutes in the kitchen'
- 'Close all the blinds in the living room → immediate execution
- 'Show me the water sensors → display of all water sensors on the dashboard
- 'Create an automation to turn off the lights at midnight → automation created
- AI has access to all the data in the house in real time
- AI learns through interactions

6.2 Intelligent Reports and Suggestions

- Automatic reports: 'Yesterday, 3 motion detections, all cameras are working, your son returned home at 6:42 PM, the oven was automatically switched off'
- AI-dictable morning report on all speakers in the house
- Proactive suggestions: 'It's raining and your office window is open. Would you like me to close it?'
- Automation suggestions based on detected patterns
- The client chooses which devices and events to include in their report

6.3 AI Infrastructure

- Lightweight AI model operating locally (Whisper for voice, local Ollama/LLM for processing)
- GPU acceleration if a graphics card is present in the server
- Voice recognition on all microphones connected to the home
- Audio response on all connected speakers
- Facial and object recognition (integrated with Frigate)
- Target latency: response in less than 2 seconds for common commands

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MODULE 7 — Cameras, NVRs & Surveillance

7.1 Automatic camera integration

- Any camera connected to the network is automatically detected and integrated (ONVIF, RTSP, HTTP).
- No manual entry of feeds or URLs — the system finds them automatically
- ASCIA PoE cameras → full profile automatically applied
- Compatible with all brands: Reolink, Dahua, Hikvision, Axis, Amcrest, etc.

7.2 Native NVR

- Continuous or event-driven recording to local storage (NAS, internal SSD)
- Navigating video history: calendar, timeline, search by event
- Automatic snapshots triggered by sensors
- Configurable retention period: 1 month to 1 year
- Storage management: disk space alerts, automatic archiving

7.3 Intelligent search within videos

The system cross-references sensor history with video recordings to enable contextual searching.

- ''Show me all the entries and exits from yesterday' → the system uses the door sensor and returns the corresponding clips
- 'Who rang the doorbell today? → Doorbell camera clips + sensor log
- Search by keywords, by area, by event type, by time

7.4 Frigate & Advanced Reconnaissance

- Native Frigate integration for object detection (people, cars, animals, packages)
- Local facial recognition (optional, with client agreement)
- Simplified interface for configuring detection zones directly on the 3D plan
- Reducing false alerts through AI validation
- Notifications with visual preview: 'A person has been detected at the entrance' + thumbnail

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MODULE 8 — Presence & Location

8.1 mmWave presence sensors (ASCIA)

ASCIA produces its own mmWave presence sensors with dual sensors: one for static detection, one for dynamic tracking.

- Static presence detection (person sitting, asleep)
- Dynamic tracking of position within the room
- Configuring zones directly from the 3D plan (delimiting by drag & drop)
- Visually configurable false alarm elimination
- Compatible with Aqara FP2 sensors and equivalent models

8.2 UWB (Ultra-Wideband) System

- Installation of UWB anchors in the house for centimeter-level positioning
- Each resident has a UWB badge or device

- Real-time location visible in the 3D map
- Automations triggered by no one: 'When Jean-Pierre enters the kitchen, turn on the coffee maker'
- Configuring UWB zones directly in the 3D interface

8.3 Bluetooth / BLE System

- UWB alternative: Bluetooth presence detection (less precise but less expensive)
- Compatible with Apple Watch, phones, BLE beacons
- Bluetooth and UWB versions are offered depending on the customer's budget.
- User profile management: height, weight, preferences related to the detected person

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MODULE 9 — Audio, Multimedia & Entertainment

9.1 Multiroom audio system

- Complete music management in all areas of the home
- Compatible with all brands of connected speakers (Sonos, Denon, Bose, Paradigm, Yamaha, ASCIA Audio (we plan to have our own amplifier, etc.)
- Sources: Spotify, Deezer, Apple Music, YouTube, web radio, local files
- Music Assistant-inspired interface with advanced controls (EQ, groups, zones)
- Voice announcements: AI can speak on all speakers
- Real-time audio synchronization between multiple rooms
- Ability to play different music in different rooms with a single Spotify account

9.2 Home Theater & Multimedia

- Simplified integration: Apple TV, Fire TV, Chromecast, Android TV, projectors
- Managing Netflix, Prime Video, Disney+, YouTube, and TV channels from the ASCIA interface
- Ability to search for and launch movies or videos from the interface
- Control multiple multimedia devices from a single interface
- 'Cinema' scene: lights dimmed, blinds down, projector on, sound configured — all in one click
- Multimedia-related automations: 'When the film starts, lower the lights to 20%'

MODULE 10 — Energy & Monitoring

10.1 Energy Dashboard

- Monitoring of consumption by device, by zone, by day/month/year
- Clear and accessible graphics for a non-technical client
- Comparison with averages and previous periods
- Estimated cost based on the customer's electricity tariff

10.2 Energy-saving automation

- Pre-designed blueprints: automatic switching off of lights when no one is present, adaptive heating, etc.
- The system automatically calculates the savings achieved
- Smart suggestions: 'Your air conditioner has been running for 8 hours while it's 18°C outside. Do you want to turn it off?'
- Detection of devices in energy-intensive standby mode

MODULE 11 — Robots & Special Devices

11.1 Robot vacuum cleaners & cleaning

- Simplified integration of all robot vacuum cleaners (Roomba, Roborock, Dreame, etc.)
- Extracting and displaying the robot map directly in the 2D/3D plane
- Zone-based start: 'Vacuum only the living room'
- Visit history, cleaning statistics
- Integration of pool robots and lawnmower robots

11.2 Special IoT Devices

- Coffee machine: scheduled start, status, coffee profiles per person
- Snow blower: remote start, snow alerts
- Pet feeder: scheduling, inventory
- Any device connected via MQTT, Zigbee, Wi-Fi or proprietary protocol

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MODULE 12 — Notifications, Reports & Events

12.1 Notification System

- Push notifications on phone, tablet, Apple Watch
- Notification with action: 'Oven on for 2 hours — Turn off / Leave on'

- Fine-grained customization: enable/disable notifications by device, by time, by user
- Notifications with integrated camera preview
- Doorbell: event with camera thumbnail + option to open the door
- Advanced notification configuration and notification creation system

12.2 Event & Reporting System

- Each event is recorded: trigger, source, time, action, result
- Security report on demand or automatically (hourly, daily, weekly)
- Morning report: dictated by AI on the speakers
- Event timeline: a chronological visualization of everything that happened
- System report: devices offline, low battery, storage, latency
- Configurable reports: the client chooses what is included.

12.3 Calendar Integration

- Connecting to Google Calendar for family members
- Automations triggered by calendar events: 'Meeting tomorrow at 7am → wake-up at 6:30am'
- Management of public holidays, birthdays, recurring events
- Smart reminder system on speakers

MODULE 13 — Data Management & Storage

13.1 Complete History

- Recording of each change of state of each device
- Automation history: who triggered what, when, and why
- Connection log: every system access recorded with location
- Configurable retention: 1 month to 1 year
- Quick search of history by device, by area, by date, by event type

13.2 Storage Management

- Real-time disk space monitoring
- Automatic alerts: 'Storage at 80% — Free up space or increase capacity'
- Automatic archiving of old data
- Supports NAS, internal SSD, and external USB storage
- Video camera management, music, media file management

13.3 Backups

- Automatic backups: daily, weekly, monthly
- Local backup + cloud option (encrypted)
- One-click restore from any backup point
- Backup success or failure notification

MODULE 14 — Diagnosis, Maintenance & Self-Healing

14.1 Self-diagnosis

- Full test that can be run at any time: lights, sensors, automations, latency, ping
- Results in the form of a report with a system health score
- Programmable: automatic daily or weekly testing
- Notification if a problem is detected

14.2 Proactive Detection

- Low battery → notification with relevant device and battery level
- Storage full → alert with release suggestion
- Device offline → notification with last known connection
- Server overheating → alert + automatic corrective action
- Home overheating → alert + ventilation suggestion
- Estimated end of life of a device → preventative notification

14.3 Self-Healing — Automatic Recovery

In case of failure or problem, the system attempts to repair itself automatically before alerting the customer or support.

- Problem detected → the system attempts one or more automatic remedies (service restart, network reconnection, configuration reload)
- If the repair is successful → log of the incident and the remedy applied
- If the problem persists → the system notifies the customer and automatically creates an ASCIA support ticket
- The ticket is sent to the ASCIA team who will contact the client for intervention.
- ASCIA support can access the system remotely with the client's agreement (temporary session).

14.4 ASCIA Telediagnosis

- With the client's agreement, the system sends a daily or weekly diagnostic report to ASCIA.
- ASCIA can intervene before the client realizes there is a problem
- Information sent: device status, batteries, network, server, errors
- The customer can deactivate remote diagnostics at any time

MODULE 15 — Interface & Customization

15.1 Responsive & multi-platform interface

- Accessible via web browser, iOS app, Android app
- Compatible with Apple Watch, tablets, and wall-mounted displays
- All features accessible from all devices
- Optimized latency: target < 50ms for local commands
- Smooth 3D interface even on mobile devices

15.2 Advanced Customization

- Full drag and drop: reposition any element of the interface
- Right-click → context menu with quick options
- Change of color, theme, layout, order of sections
- Dedicated editing environment: 'Edit dashboard' mode
- Configurable keyboard shortcuts
- Favorites and custom shortcuts at the top of the interface

15.3 Update Management

- Manual update: the customer decides when to update
- Programmable automatic update: weekly, monthly, annual
- Detailed update notes before installation
- Rollback possible in case of problems after update

MODULE 16 — Voice Assistants & External Integrations

16.1 Google Home & Amazon Alexa

- Native and simplified integration of Google Home and Amazon Alexa

- Exposing ASCIA devices to Google/Alexa in just a few clicks
- Creating custom phrases for voice commands
- Alexa routines created by the client are translated directly into ASCIA OS
- ASCIA automations are accessible via Google Home
- Bilingual management: French and English natively supported

16.2 Journey Management & Mobility

- The customer registers their usual addresses (work, school, etc.)
- Waze/Google Maps API integration: real-time travel time estimation
- 'You have a meeting at 8am, the journey takes 35 minutes with the current traffic — I'll wake you up at 7am.
- Departure automations: 'When Pierre leaves work → preheat the house'
- Return notification: 'Jean-Pierre will arrive in 10 minutes → prepare the house'

MODULE 17 — Nomenclature & Automatic Organization

17.1 Automatic Intelligent Naming

The system automatically names devices and entities in a consistent, intuitive, and easy-to-remember way, based on the initial submission made to the client.

- The ASCIA submission already contains the names of the rooms and zones → the system uses this data for automatic naming
- Example: door sensor in 'Bedroom 1' → named 'Bedroom 1 Door Sensor'
- Consistent naming conventions throughout the interface: dashboard, automations, voice commands
- Complete list of all names, exportable and editable
- Modifications can be made at any time with automatic propagation throughout the system.

17.2 Automatic Zone Assignment

- When a device is added, the system asks for its zone of origin.
- The device is automatically placed in the correct area of the 3D plane.
- All dashboards, groups, and automations related to this area are updated.
- Delineate areas directly in the 3D model by drawing or drag and drop.

MODULE 18 — ASCIA Business Process & Deployment

18.1 Business Workflow

The ASCIA business process is fully integrated with the home automation system to maximize efficiency and minimize errors.

Step 1 — On-site submission

- The ASCIA technician visits the customer's premises with the ASCIA POS system.
- The submission lists all the client's devices, zones, and needs.
- The names of the parts and areas are entered upon submission.
- Example: Living room presence sensor: sensor.presence_RDC_salon_1

Step 2 — Transmission & Logistics

- The validated submission is sent to the logistics manager
- Automatic ordering of necessary parts
- The ASCIA server is pre-configured with the submission data.

Step 3 — Installation

- The client receives: ASCIA server + PoE switches + devices
- Simple connection: modem → switch → server + devices
- Accessing the ASCIA interface from a phone or computer
- Initial guided setup with startup wizard

Step 4 — Client Configuration

- Uploading the 3D plan or creating it in the interface
- The devices automatically appear in the system
- The customer assigns, names, and positions their devices in minutes.
- The basic automations are already ready to activate.

18.2 Continuous Support

- Proactive remote diagnostics: ASCIA detects problems before the customer
- Temporary support access with customer agreement
- Automatic support ticket generated in case of unresolved issues.
- Remote firmware update for ASCIA devices